

**Transformations of  $f(x) = |x|$ , Part II**

Algebra II Honors | Unit 2 | Day 3 | TEK A2.6.C

**Objective:** I can derive why  $(c, d)$  is always the vertex of  $f(x) = a|x - c| + d$ , and work backward from a vertex and a point to find the equation.

**Vocabulary**

**Horizontal shift** —  $f(x - c)$ ; shifts opposite the sign of  $c$ .

**Vertical shift** —  $f(x) + d$ ; shifts the same direction as the sign of  $d$ .

**Vertex form** —  $f(x) = a|x - c| + d$ , written specifically to make the vertex \_\_\_\_\_ easy to read off.

**Part 1 — Prove the Vertex Is Always  $(c, d)$** 

Evaluate  $f(x) = a|x - c| + d$  at  $x = c$ , for any  $a, c, d$ :

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$$f(c) = a|c - c| + d = a|0| + d = a \cdot \underline{\hspace{1cm}} + d = \underline{\hspace{1cm}} \rightarrow \text{so } (c, d) \text{ is always ON the graph}$$

Explain why  $(c, d)$  is the LOWEST point when  $a > 0$  (or highest when  $a < 0$ ), not just A point on the graph.

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Sample: \_\_\_\_\_

**Part 2 — Working Backward: Vertex + One Point  $\rightarrow$  Equation**

*A function has vertex  $(3, -2)$  and passes through  $(5, 4)$ . Find  $a$ , then write the equation.*

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$$a|5 - 3| + (-2) = 4 \rightarrow a(2) - 2 = 4 \rightarrow 2a = \underline{\hspace{1cm}} \rightarrow a = \underline{\hspace{1cm}}$$

$$\text{Equation: } f(x) = \underline{\hspace{2cm}}$$

You Try. Vertex  $(-1, 5)$ , passes through  $(2, -4)$ . Find  $a$  and write the equation.

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$$a|2 - (-1)| + 5 = -4 \rightarrow 3a = \underline{\hspace{1cm}} \rightarrow a = \underline{\hspace{1cm}} \quad f(x) = \underline{\hspace{2cm}}$$

A second one. Vertex  $(4, 1)$ , passes through  $(7, 10)$ . Find  $a$  and write the equation.

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$$a|7 - 4| + 1 = 10 \rightarrow 3a = \underline{\hspace{1cm}} \rightarrow a = \underline{\hspace{1cm}} \quad f(x) = \underline{\hspace{2cm}}$$

**Part 3 — Connecting Back to Day 2's Factoring**

$f(x) = 2|3x - 12| + 5$  has a  $b$ -value hiding inside. Factor to find the vertex form  $a|x - c| + d$ .

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$$2|3x - 12| + 5 = 2|3(x - 4)| + 5 = 2 \cdot \underline{\hspace{1cm}} \cdot |x - 4| + 5 = \underline{\hspace{1cm}}|x - 4| + 5 \rightarrow \text{vertex } \underline{\hspace{2cm}}$$

## Part 4 — Real World: Dunhill Drafting's Roofline

Roofline data was modeled by trial-and-error transformation, landing on  $f(x) = -1.2|x - 3| + 8$ .

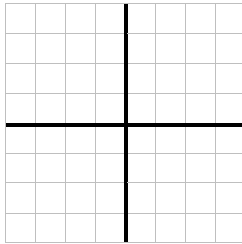
Vertex (highest point of the roof): \_\_\_\_\_ Explain how you know it is a MAXIMUM, not a minimum, without graphing.

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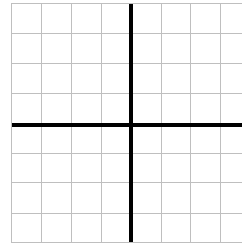
## Part 5 — Graph and Compare

Graph both. Each square is 1 unit; both axes run -5 to 5.

A.  $f(x) = |x - 2| - 3$



B.  $f(x) = |x + 3| + 2$



## Practice

1. Give the vertex of  $f(x) = -3|x + 8| - 6$ . \_\_\_\_\_
2. Factor  $f(x) = 3|2x + 6| - 1$  into vertex form and give the vertex. \_\_\_\_\_ vertex  
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3. A function has vertex  $(-2, 8)$  and passes through  $(1, -1)$ . Find  $a$  and write the equation.  
 $a|1 - (-2)| + 8 = -1 \rightarrow 3a = \underline{\hspace{2cm}} \rightarrow a = \underline{\hspace{2cm}} \quad f(x) = \underline{\hspace{4cm}}$
4. Factor  $f(x) = 5|10x - 30| + 2$  into vertex form, and give both the vertex and the real  $a$ -value.  
\_\_\_\_\_ vertex \_\_\_\_\_  $a = \underline{\hspace{2cm}}$

## Exit Ticket

A function has vertex  $(2, -5)$  and passes through  $(0, -1)$ . Find  $a$  and write the equation, then explain how you know  $(2, -5)$  is a minimum.

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$a|0 - 2| + (-5) = -1 \rightarrow a = \underline{\hspace{2cm}} \quad f(x) = \underline{\hspace{4cm}}$

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